

Spinal fracture lesions segmentation based on U-net

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Abstract—Because of the problem that the complexity of spine CT images, the irregular shape of vertebral boundary, low contrast, noise and unevenness in images, meanwhile there are artificial deviations and low efficiencies in clinic, which needs doctors' prior knowledge and clinical experience to determine lesions location in CT images, so it can not meet the clinical real-time needs. In this paper, We use deep learning to process the CT images of spine, and to divide lesions of (cervical fracture, cfracture), (thoracic fracture, tfraction), (lumbar fracture, lfraction) by the improved U-net^[1]. The experiment shows that we can effectively segment spinal fracture lesions by U-net, which can basically meet the clinical real-time needs.

Keywords—Deep learning U-net Segementation

I. INTRODUCTION

At present, spondylosis has become a common and high incidence, and because of the long-term use of computers, watching TV and other factors, spondylosis has a trend of young age. According to epidemiological data, in China, about 40% of people over 40 years old have different spinal health problems. In the United States, more than 25% of the occupational diseases are caused by the problem of the spine. More than 250000 lumbar surgeries are carried out in the United States every year, which costs 6 billion US dollars.

Because of the complexity of spine image, so far, there is no satisfactory separation method for all spine images. Combined with the literature and clinical practice, it is found that the current spine image separation methods are either for a certain type of image, or the accuracy and speed are difficult to balance development, and the degree of automation is low. In addition, most of the spine separation images that can be used in clinic are CT and X-ray, mainly for the diagnosis of bone diseases, and can not accurately solve the problem of image drift during the operation. During the operation, the tissue structure will shift, making the image and the real position have errors, according to statistics, the error rate is as high as 66%^[2]. At present, more mature and classic spine image separation methods include: Watershed Based spine separation^[3], horizontal set based spine separation^[4-6], model-based spine separation^[7-9], Atlas Based spine separation and other separation methods.

In this paper, we use a U-type convolutional neural network (U-net) to segment spinal lesions. Firstly, Gauss and Laplace filtering is used to enhance the details of the original CT image, and then U-net is used to segment the

preprocessed image. The results of qualitative and quantitative experiments show that the segmentation method based on U-net can effectively segment spinal lesion.

II. SPINAL FRACTURE LESION SEGMENTATION BASED ON U-NET

We use U-net to process spinal fracture lesions in this paper, as shown in figure 1. The traditional U-net algorithm does not process the original CT image. In this paper, the original image is processed by Laplace transform and output to Python readable image format. Secondly, the preprocessed image is trained by U-net to get the segmentation model, calculate the loss function optimization model, and use the test set to test the trained model. Finally, the accuracy of the segmentation algorithm is evaluated quantitatively.

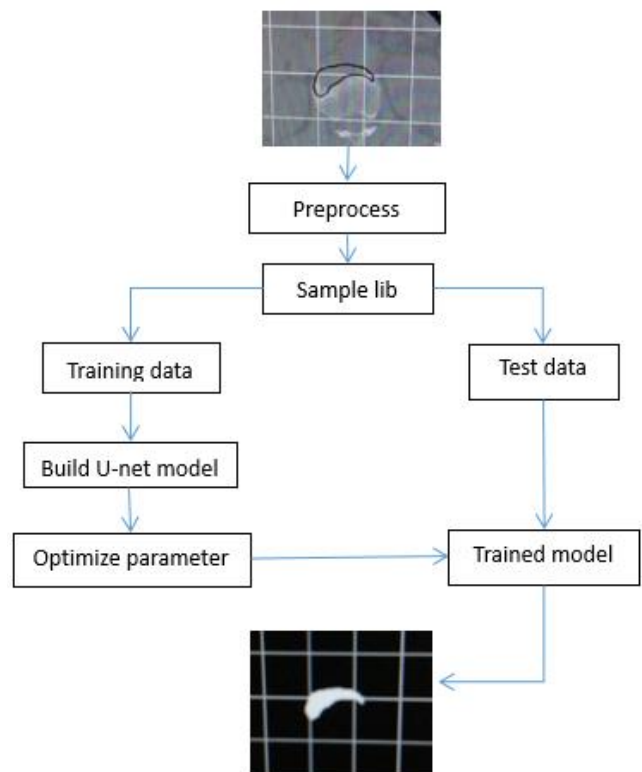


Figure 1 overall framework of spinal fracture lesions segmentation

A. Preprocess CT images

In order to construct the training set and test set for u-net, remove the peripheral useless data, only keep the

region of interest needed by the training network, and preprocess the original data. Figure 2 shows the data preprocessing process. First of all, we use matlab to preprocess DICOM format image. Considering all kinds of processing methods, we choose Gauss filter and mathematical morphology to process CT image. Because this paper uses the keras framework for experiments and uses Python to write code, it outputs Python readable image format.

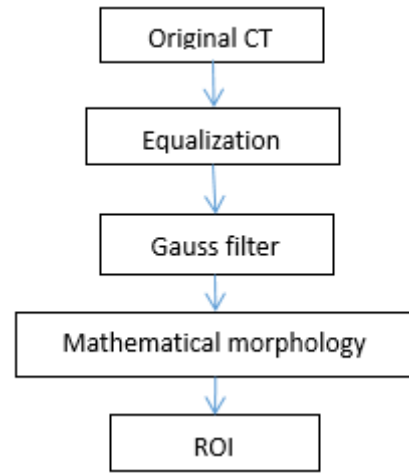


Figure 2 preprocess CT images

B. Segmentation based on U-net

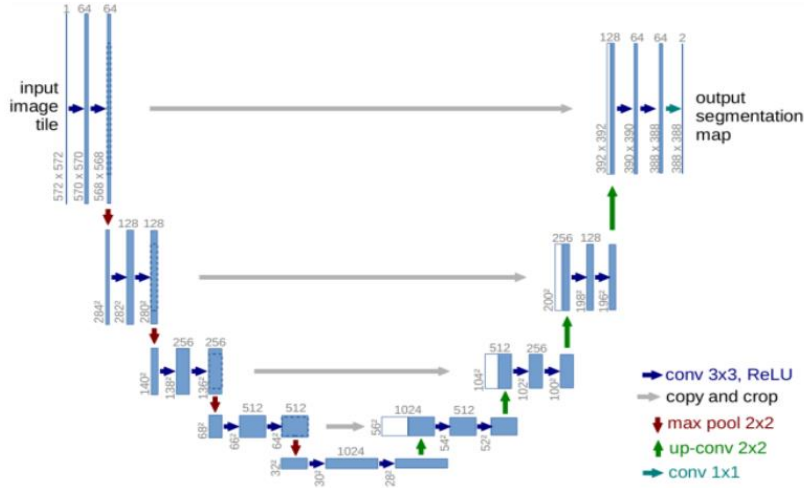


Figure 3 U-net architecture

U-net is a 2D full convolution neural network, Figure 3 shows the u-net network structure. U-net has a symmetrical U-shaped network structure, and the left side is a contraction path, right side is expansion path. The compression structure adopts convolution operation and pooling operation for feature extraction, and the structure repeatedly adopts two 3×3 convolutions and one 2×2 maximum pooling. The expansion structure uses deconvolution and up sampling to restore the image size to classify each pixel. In the structure, eight 3×3 convolutions, one 1×1 convolution and four 2×2 up sampling are used.

The u-net model adopts the idea of encoder decoder. The left compression structure is the encoder, the right expansion structure is the decoder, and the encoder decoder. The idea of coder is widely used in deep learning semantic segmentation model.

III. EXPERIMENTS AND ANALYSIS

A. Experimental data

The experimental data in this paper are accumulated clinical data from Xijing Hospital, the Second Affiliated Hospital of Shaanxi University of Traditional Chinese Medicine and other hospitals. We use the software labelme

to mark the location, size and category of the lesion in each CT image, and classify as (cervical fracture, cfraction), (thoracic fracture, tfraction), (lumbar fracture, lfraction) according to lesion category. .

We get 40 cases spinal fracture clinical data from hospital, which including 5134 CT images, it is impossible to include all kinds of spinal fracture lesions. So we add data by data augmentation which rotating the image 180 degrees, and flipping left and right, up and down. After data enhancement, the data set of spinal focus was expanded to 10268 CT images.

We randomly generates the testing set according to 15%, In training verification set, 85% are used as training set, and the remaining 15% are used as verification set. So 4363 CT images are selected for training sets and 771 CT images are test set.

B. Experimental environment

We use CPU: Intel Core i7-6700 @ 3.40GHz * 8, memory 32G, graphics card GTX1070, 8G, 128G + 1t hard disk, operating system is windows 10, python 3.5, Cuda8.0 and Cudnn6.0 to train and test.

C. Experiment

1) Experiments results

The experiments results are shown in figure 4, a,c,e is lfracture,tfracture,cfracture,and b,d,f is pridect segmentation lesions. The average segmentation accuracy is 0.885, which is slightly better than the current commonly used segmentation methods.

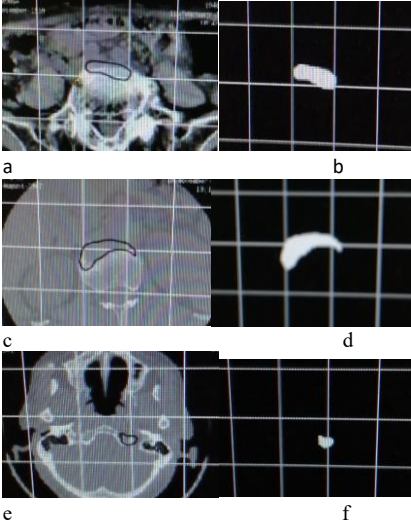


Figure 4 Experiments results

In this paper, JS index is used to evaluate the accuracy of the proposed segmentation algorithm. Eq. (1) is the definition of JS. S1 is the result of manual annotation and S2 is the result of segmentation.

$$J(s_1, s_2) = \frac{|s_1 \cap s_2|}{|s_1 \cup s_2|} \quad (1)$$

2) Experiments results analysis

In order to evaluate the results, this paper uses the accuracy measurement method to evaluate the u-net segmentation results. The calculation formula of accuracy (AC) is shown as Eq.(2):

$$AC = \frac{TP + TN}{TP + FP + TN + FN} \quad (2)$$

TABLE I SEGMENTATION RESULTS

True	Test	
	Lesion	not lesion
Lesion	0.867	0.133
Not lesion	0.108	0.892

The segmentation results is shown in Table I, accuracy of model is 87.9%, the misclassification rate was 12.1%, false positive rate was 10.8%, false negative rate was 13.3%.

The data show that u-net has a good effect on the segmentation of spinal lesions, and it can play an auxiliary role in the diagnosis of most patients.

IV. CONCLUSION

Based on the u-net deep learning network model, this study uses the actual clinical CT image data set as training data to realize the segmentation of spine fracture focus. The main feature is to use the deep neural network learning feature to automatically segment spine focus. From the experimental results, we can see that this method can effectively segment the spinal lesions. However, there are more false positive and false negative spinal lesions in the segmentation results, so improving the false positive and false negative segmentation results is the next research focus.

Because the deep learning procession of medical image needs a lot of data, but most of the medical image data sets are labeled manually at present, so the workload is large. In the future, we will further explore whether the medical image data sets can be automatically labeled to reduce the workload, so as to better solve the clinical problems of the hospital.

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